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To: Editor-in-Chief / Handling Editor
Journal of the Atmospheric Sciences
American Meteorological Society (AMS)

Subject: Submission of original research manuscript entitled “*VORTEX 6.0: A Low-Cost Kinematic Reduced-Order Discrete Vortex Model with Recursive Sub-Grid Fractal Turbulence Cascade for Real-Time Atmospheric Dynamics*”

Dear Editor,

We are pleased to submit our original research article entitled “**VORTEX 6.0: A Low-Cost Kinematic Reduced-Order Discrete Vortex Model with Recursive Sub-Grid Fractal Turbulence Cascade for Real-Time Atmospheric Dynamics**” for publication consideration as a Research Article in the *Journal of the Atmospheric Sciences*.

Microscale tornadic and mesocyclonic dynamics present a fundamental computational challenge in atmospheric science: full three-dimensional Navier-Stokes solvers (such as WRF and Large-Eddy Simulations) offer high physical fidelity but require prohibitive computational resources ($\mathcal{O}(10^3 - 10^5)$ CPU hours per run). This latency prevents their use in instantaneous multi-scenario ensemble forecasting and real-time emergency decision support.

In this work, we present **VORTEX 6.0**, a kinematic Reduced-Order Discrete Vortex Model (ROM-DVM) designed to bridge the operational gap between computational speed and physical realism. Key scientific and technical contributions of our study include:

- **Recursive Sub-Grid Cascade:** A novel fractal branching algorithm that emulates the forward transfer of turbulent kinetic energy (TKE) mirroring Kolmogorov $k^{-5/3}$ scale invariance down to micro-scales.
- **Vortex Interaction Mechanics:** Formulation of continuous core dissipation, shear strain energy decay, co-rotating merger dynamics, and alignment repulsion for interacting vortex elements.
- **Unprecedented Computational Efficiency:** Benchmark results demonstrating integration speeds of 0.108 s per 100 steps on commodity hardware—over four orders of magnitude faster than conventional grid-based CFD solvers.

Given recent spatial and temporal shifts in U.S. tornadic climatology—characterized by increased tornadic activity in the Mid-South and Midwest regions—there is an urgent need for surrogate modeling tools capable of rapid ensemble screening. We demonstrate that VORTEX 6.0 provides a robust, complementary framework to full-physics NWP models, enabling instant trajectory risk assessment and physical scenario exploration.

We confirm that:

1. This manuscript is an original work, has not been published previously, and is not currently under consideration for publication by any other journal.
2. All listed authors have reviewed and approved the final manuscript for submission.
3. The authors declare no competing financial or personal interests.

Thank you very much for your time, consideration, and management of our submission.
We look forward to hearing from you regarding the peer-review process.

Sincerely,

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